Parametric catalogs: thousands of near-identical work items

BC3 FIEBDC

47,513 items
25 parametric templates
ADIF railway catalog, subcategory OEB

Template OEB040$: Canalización hormigonada de tubo de polietileno 160mm

ID	Nº tubos	Terreno	Trabajo
OEB040aabcb	1	Normal	Nocturno
OEB040abbcb	1	Bajo vías	Nocturno
OEB040babcb	2	Normal	Nocturno
OEB040bbbcb	2	Bajo vías	Nocturno

11 × 8 × 6 × 4 × 3 = 6,336 variants of one template, differing only in parameter values

Nº tubos (11) · Terreno (8) · Trabajo (6) · Banda (4) · Condiciones (3)

142 retrieval configurations, from lexical to structured, each scored at two levels

Parse BC3 → Add parameter tokens → Retrieve top 100 → Score at item and parent level

Test query q_003907

canalización hormigonada 1 t, polietileno libre de halógenos de 160 mm, normal. (nocturno/i < "3" horas/volumen escaso)

Parameter tokens

- n_tubos_1
- terreno_normal
- trabajo_nocturno
- banda_i_lt_3_horas
- condic_volumen_escaso

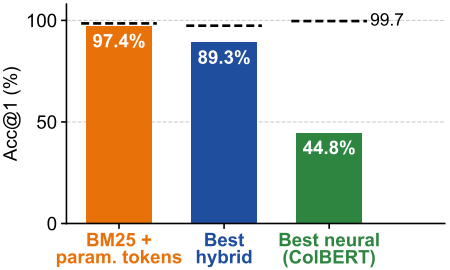
BM25 top 3 (score)

1. OEB040aabcb — 29.3
   1 t · i < 3 horas
2. OEB040aabbb — 27.3
   1 t · 3 <= i < 5 horas
3. OEB040aabab — 27.0
   1 t · i >= 5 horas

✓ item level ✓ parent level

Dense models find the family, not the exact variant

Bars: exact variant · dashed: right family



Numeric mismatches: 97.5–98.2% of errors

in BM25, BGE-M3 ColBERT and the hybrid alike

Maintenance window (BGE-M3 ColBERT)

3 <= i < 5 horas vs i < 3 horas

Number of tubes (BM25 unigram)

5 tubes vs 3 tubes

retrieved correct